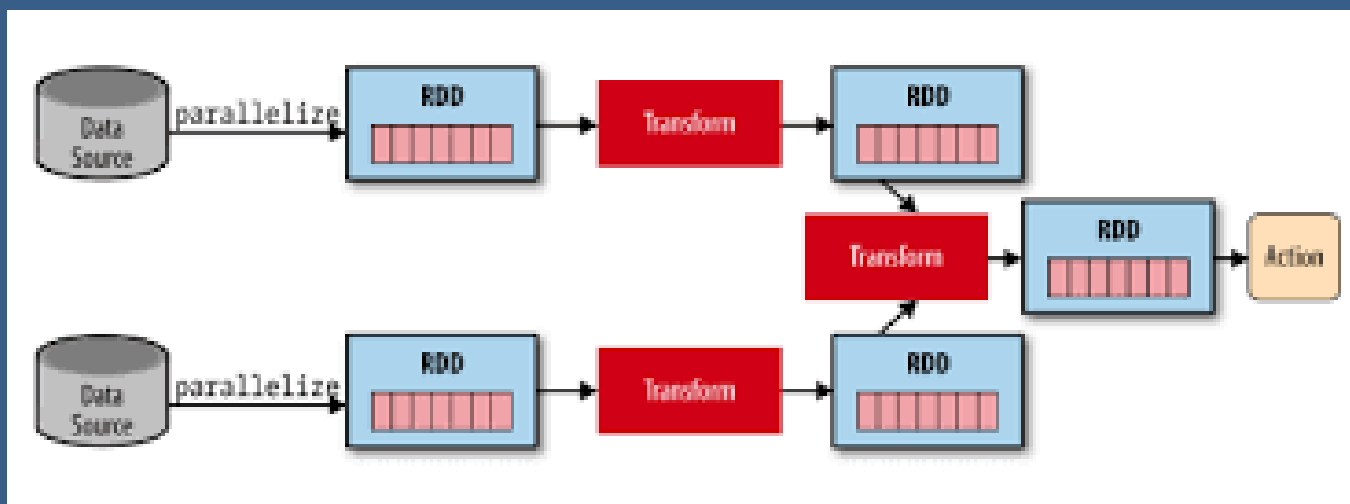




Resilient Distributed Datasets (RDDs)

The Foundation of Apache Spark

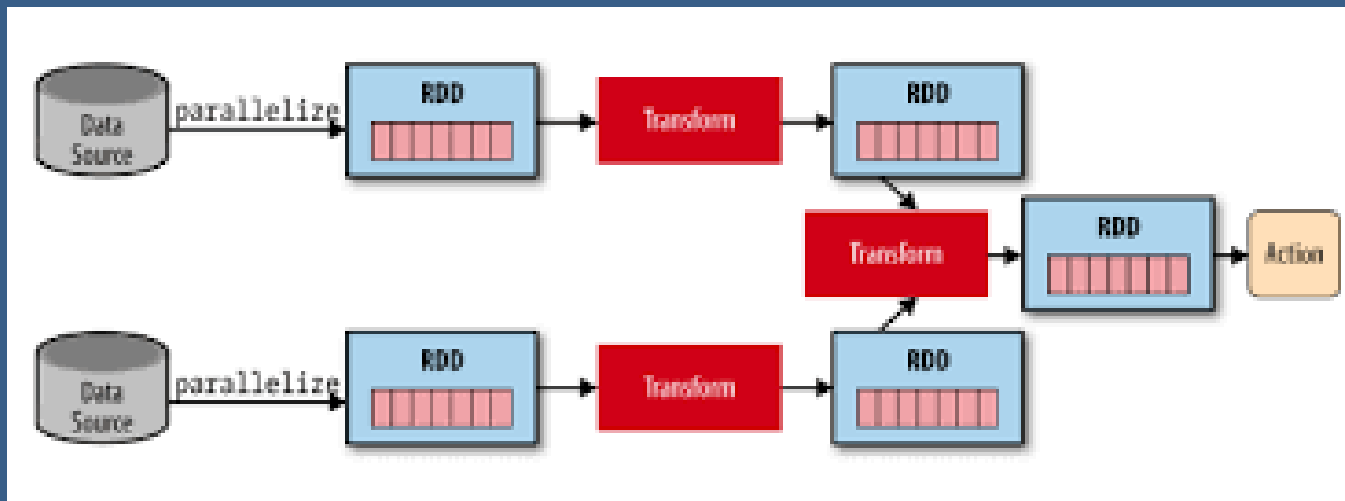
WHAT IS RDD?



- RDDs is the primary data abstraction in Apache Spark and the core of Spark that we often refer to as “Spark Core”.
- A RDD is a resilient and distributed collection of records spread over one or many partitions
- One could compare RDDs to collections in Scala, i.e. a RDD is computed on many JVMs while a Scala collection lives on a single JVM.

WHAT IS RDD?

- RDDs are a container of instructions on how to materialize big (arrays of) distributed data, and how to split it into partitions so Spark (using [executors](#)) can hold some of them.



RDD's are the Fundamental unit of data in spark

An RDD belongs to one and only SparkContext. You can not share RDD's between contexts.



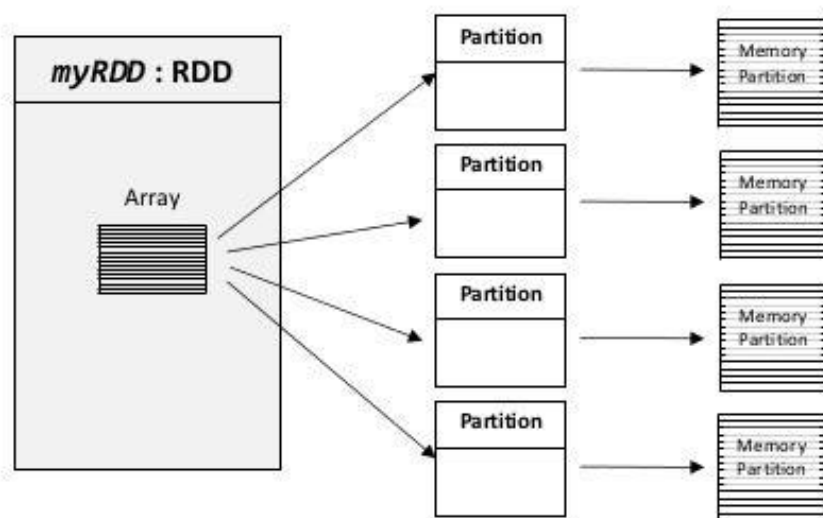
Features of RDD'S?

• **Resilient**, i.e. fault-tolerant with the help of RDD lineage graph and so able to recompute missing or damaged partitions due to node failures.

• **Distributed** with data residing on multiple nodes in a cluster.

• **Dataset** is a collection of partitioned data with primitive values or values of values, e.g. tuples or other objects (python, scala or java)

What is an RDD?



Some RDD Characteristics

- Hold references to Partition objects
- Each Partition object references a subset of your data
- Partitions are assigned to nodes on your cluster
- Each partition/split will be in RAM (by default)

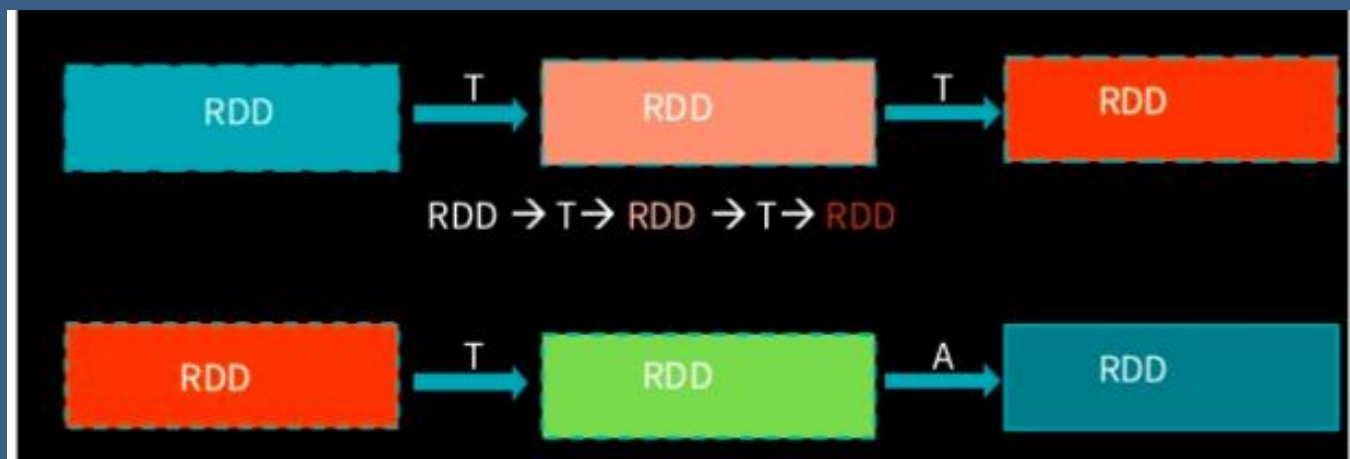
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FEATURES OF RDD'S?

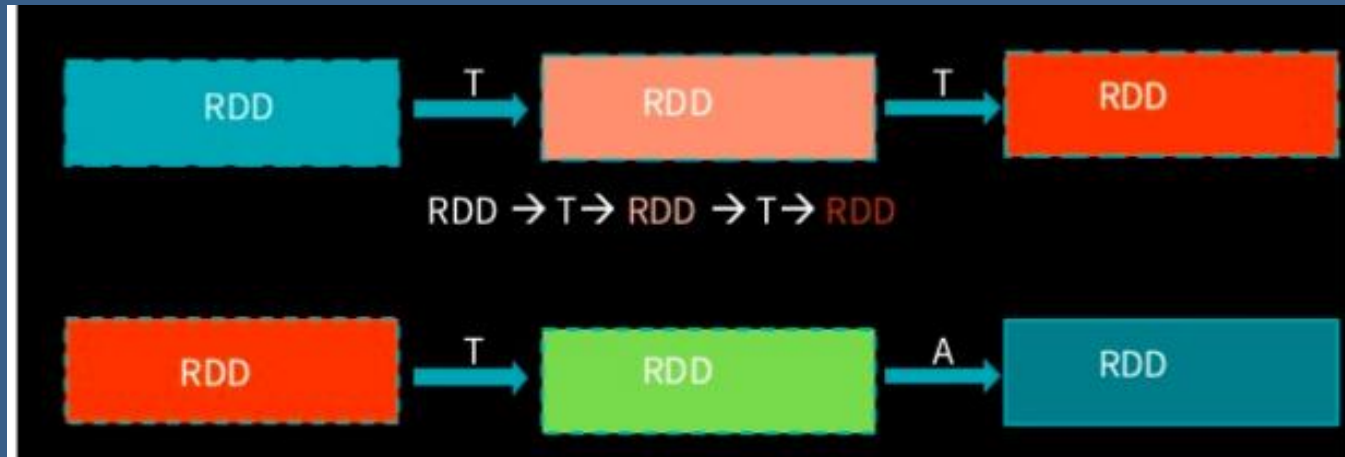
- **In-Memory**, i.e. data inside RDD is stored in memory as much (size) and long (time) as possible.
- **Immutable** or **Read-Only**, i.e. it does not change once created and can only be transformed using transformations to new RDDs.
- **Lazy evaluated** i.e. transformation on RDD, don't get performed immediately. Spark internally records the metadata to track the operations. Lazy evaluation reduce the number of passes on the data by grouping the operations.



T-Transform,
A- Action



FEATURES OF RDD'S?



- **Cacheable**, i.e. you can hold all the data in a persistent “storage” like memory (default and the most preferred) or disk (the least preferred due to access speed).
- **Parallel**, i.e. process data in parallel.
- **Partitioned** – records are partitioned (split into logical partitions) and distributed across nodes in a cluster.
- **Location Stickiness**- RDD can define placement preference to compute partitions